EXP 6

Exploratory Data Visualization

AIM:

To understand the patterns in the student performance dataset by drawing a line chart, a bar chart and a histogram, and to study how study time affects the marks and how the number of absences is distributed.

ALGORITHM:

1. Import pandas, matplotlib and numpy.
2. Load the student performance file (stu_per.csv) into a DataFrame and display the first five rows.
3. Group the data by studytime and calculate the average of the marks G1, G2 and G3.
4. Draw a line chart of the average G1, G2 and G3 marks for each study time.
5. Draw a bar chart of the average final mark (G3) for each study time.
6. Draw a histogram of the number of absences of the students.
7. Observe the charts and write the result.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np



df = pd.read_csv("stu_per.csv")

print(df.head())
```

```
# Group by study time and calculate average scores
grouped_data = df.groupby("studytime")[["G1", "G2", "G3"]].mean()

print(grouped_data)

plt.figure(figsize=(8, 5))

plt.plot(
    grouped_data.index,
    grouped_data["G1"],
    marker="o",
    label="G1"
)

plt.plot(
    grouped_data.index,
    grouped_data["G2"],
    marker="s",
    label="G2"
)
```

```
plt.plot(
    grouped_data.index,
    grouped_data["G3"],
    marker="^",
    label="G3"
)

plt.title("Study Time vs Student Performance")
plt.xlabel("Study Time")
plt.ylabel("Average Score")
plt.legend()
plt.grid(True)

plt.show()
```

```
study_score = df.groupby("studytime")["G3"].mean()

plt.figure(figsize=(8, 5))

plt.bar(
    study_score.index,
    study_score.values
)

plt.title("Average Final Score by Study Time")
plt.xlabel("Study Time")
plt.ylabel("Average G3 Score")

plt.show()
```

```
plt.figure(figsize=(8, 5))

plt.hist(
    df["absences"],
    bins=10,
    edgecolor="black"
)

plt.title("Distribution of Student Absences")
plt.xlabel("Number of Absences")
plt.ylabel("Number of Students")

plt.show()
```

OUTPUT:

Output of: import pandas as pd

```
  school sex  age address famsize Pstatus  Medu  Fedu      Mjob      Fjob  ...  \
0     GP   F   18       U     GT3       A     4     4  at_home   teacher  ...
1     GP   F   17       U     GT3       T     1     1  at_home     other  ...
2     GP   F   15       U     LE3       T     1     1  at_home     other  ...
3     GP   F   15       U     GT3       T     4     2   health  services  ...
4     GP   F   16       U     GT3       T     3     3    other     other  ...

   famrel freetime  goout  Dalc  Walc health absences  G1  G2  G3
0       4        3      4     1     1      3        4   0  11  11
```

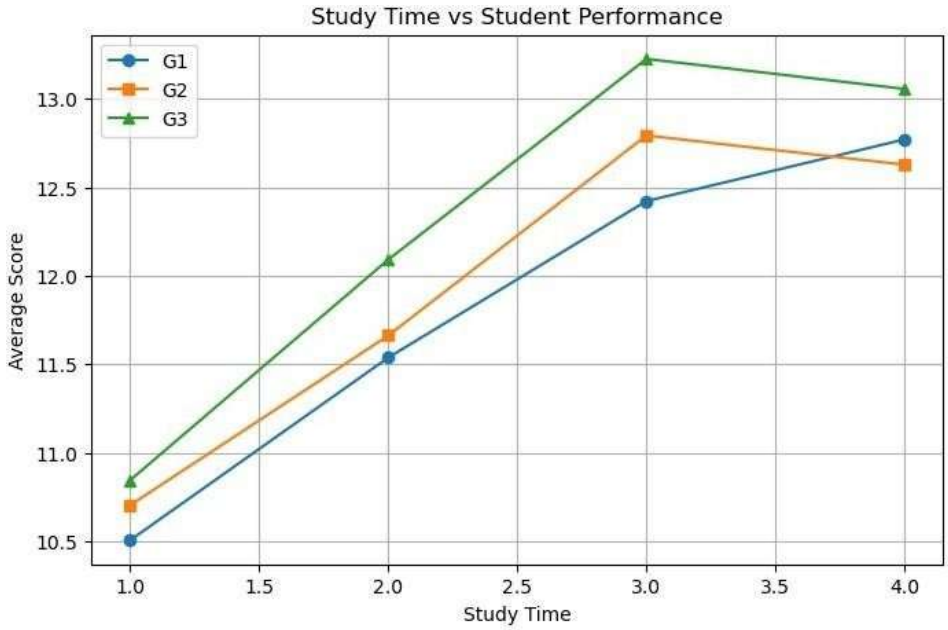
```
1       5         3       3      1      1       3         2    9  11  11
2       4         3       2      2      3       3         6   12  13  12
3       3         2       2      1      1       5         0   14  14  14
4       4         3       2      1      2       5         0   11  13  13

[5 rows x 33 columns]
```

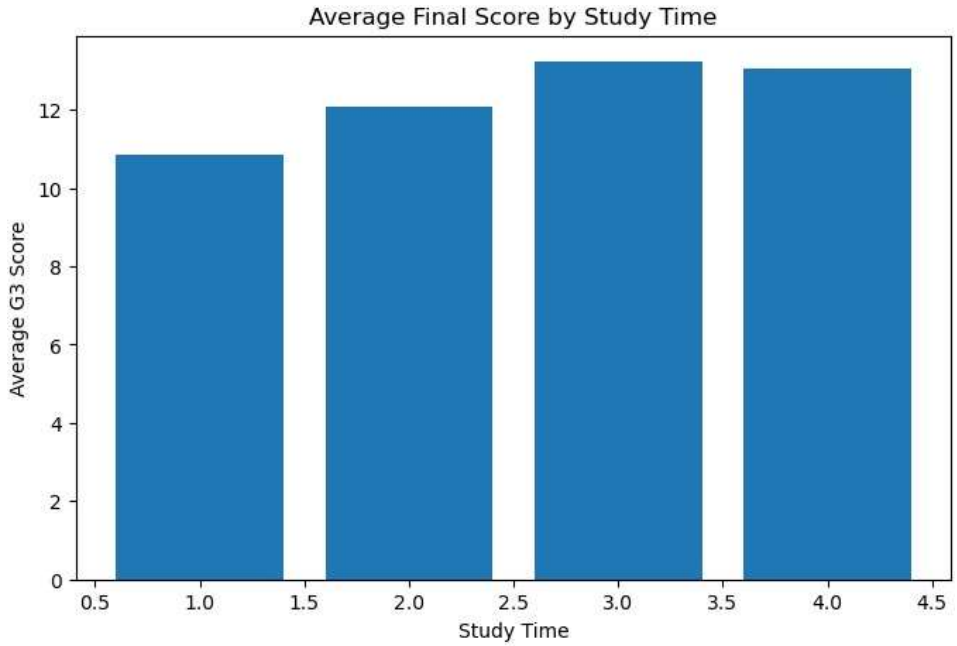
Output of: grouped_data = df.groupby("studytime")[["G1", "G2", "G3"]].mean()

```
                 G1         G2         G3
studytime
1          10.504717  10.702830  10.844340
2          11.537705  11.662295  12.091803
3          12.422680  12.793814  13.226804
4          12.771429  12.628571  13.057143
```

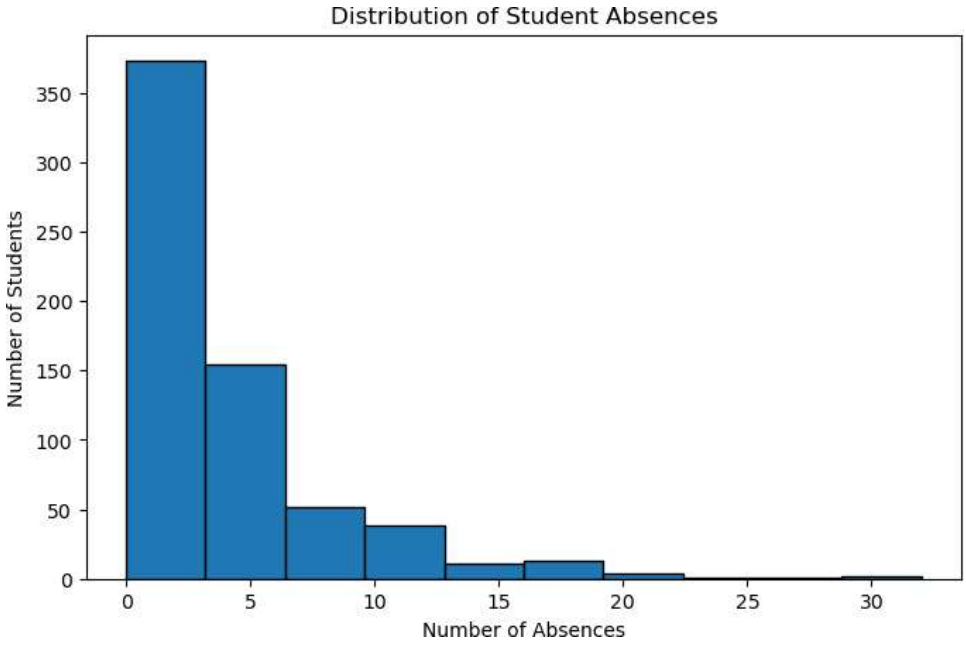
Output of: grouped_data = df.groupby("studytime")[["G1", "G2", "G3"]].mean()



Output of: study_score = df.groupby("studytime")["G3"].mean()



Output of: plt.figure(figsize=(8, 5))



RESULT:

Thus, the student performance data was visualized using a line chart, a bar chart and a histogram. The average final mark (G3) increased from 10.84 for study time 1 to 13.23 for study time 3, showing that students who study more tend to score higher marks.